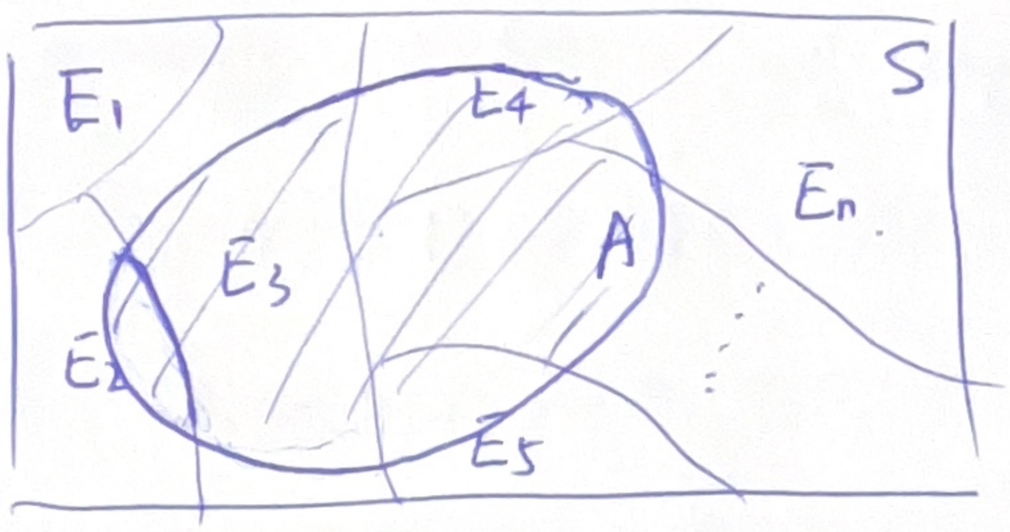
Law of total probability

Suppose events $E_1, E_2, \dots E_n$ form a partition. i.e.,

1) $E_i \cap E_j = \phi \quad \forall\, i \neq j$

2) $E_1 \cup E_2 \cup \dots \cup E_n = S$



Then for any event A

$$P(A) = P[(E_1 \cap A) \cup (E_2 \cap A) \cup \dots \cup (E_n \cap A)]$$

$$\overset{m.e}{=} P(E_1 \cap A) + P(E_2 \cap A) + \dots + P(E_n \cap A)$$

$$= P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + \dots + P(E_n)P(A|E_n)$$

Baye's Rule

Suppose $E_1, \dots E_n$ form a partition.

then for any event A $(P(A) \neq 0)$

$$P(E_i|A) = \frac{P(A \cap E_i)}{P(A)}$$

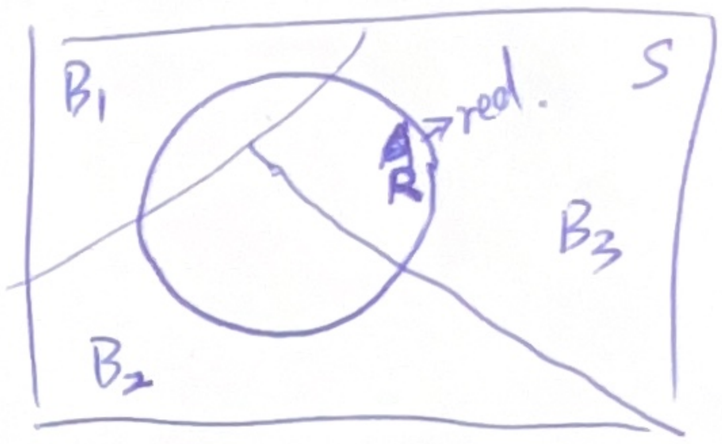
$$= \frac{P(E_i)P(A|E_i)}{P(E_1)P(A|E_1) + \dots + P(E_n)P(A|E_n)}$$

Example: 3 bowls.

$B_1 \rightarrow$ 1 red 4 white chips.

$B_2 \rightarrow$ 2 red 1 white chips.

$B_3 \rightarrow$ 4 red 1 white . . .



randomly select. one bowl, then randomly select a chip from that bowl.

a) Find probability to get a <u>red</u> chip.

$$P(R) = P[(B_1 \cap R) \cup (B_2 \cap R) \cup (B_3 \cap R)]$$

$$\overset{\text{me.}}{=} P(B_1 \cap R) + P(B_2 \cap R) + P(B_3 \cap R)$$

$$= P(B_1)P(R|B_1) + P(B_2)P(R|B_2) + P(B_3)P(R|B_3)$$

$$= \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{2}{3} + \frac{1}{3} \times \frac{4}{5}$$

$$= \frac{5}{9}$$

b) <u>Given</u> the chip was R., find the probability of B_3.

$$P(B_3|R) = \frac{P(B_3 \cap R)}{P(R)} = \frac{P(B_3)P(R|B_3)}{P(B_1)P(R|B_1) + P(B_2)P(R|B_2) + \underline{P(B_3)P(R|B_3)}}$$

$$= \frac{\frac{1}{3} \times \frac{4}{5}}{\frac{5}{9}} = 0.48.$$

c) $$P(B_2|R) = \frac{P(B_2 \cap R)}{P(R)} = \frac{P(B_2)P(R|B_2)}{P(R)}$$

$$= \frac{\frac{1}{3} \times \frac{2}{3}}{\frac{5}{9}} = 0.4$$

$$P(B_1 | R) = 1 - P(B_1' | R)$$

$$= 1 - P(B_2 | R) - P(B_3 | R)$$

$$= 1 - 0.48 - 0.48 = 0.12$$

$P(B_1)$, $P(B_2)$, $P(B_3)$ $\rightsquigarrow$ prior probabilities.

$P(B_1|R)$, $P(B_2|R)$, $P(B_3|R)$ $\rightsquigarrow$ posterior probabilities.

<u>Example</u>: A Boy has 5 black 4 white chips in the left pocket.
4 black 5 white chips in the right pocket.

① R.S. a chip from L and transfer it to R.

② R.S a chip from R.

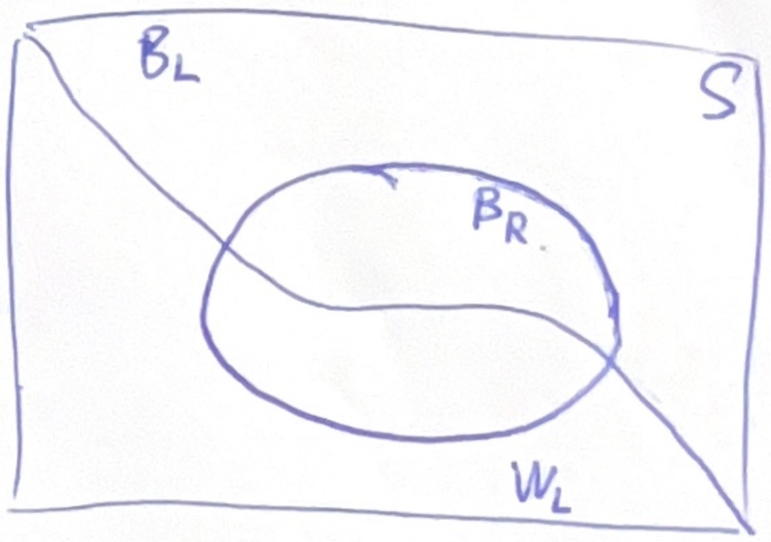
a) Find the probability of black chip.

$B_L \rightarrow$ B from L.

$W_L \rightarrow$ W from L.

$B_R \rightarrow$ B from R.

$W_R \rightarrow$ W from R.



$$P(B_R) = P[(B_L \cap B_R) \cup (W_L \cap B_R)]$$

$$\overset{m.e.}{=} P(B_L \cap B_R) + P(W_L \cap B_R)$$

$$= P(B_L)P(B_R|B_L) + P(W_L)P(B_R|W_L)$$

$$= \frac{5}{9} \times \frac{5}{10} + \frac{4}{9} \times \frac{4}{10} = \frac{41}{90}$$

b> $P(B_L \mid B_R) = \dfrac{P(B_L \cap B_R)}{P(B_R)} = \dfrac{P(B_L)P(B_R \mid B_L)}{P(B_R)}$

$$= \frac{\frac{5}{9} \times \frac{5}{10}}{\frac{41}{90}} = \frac{25}{41}$$

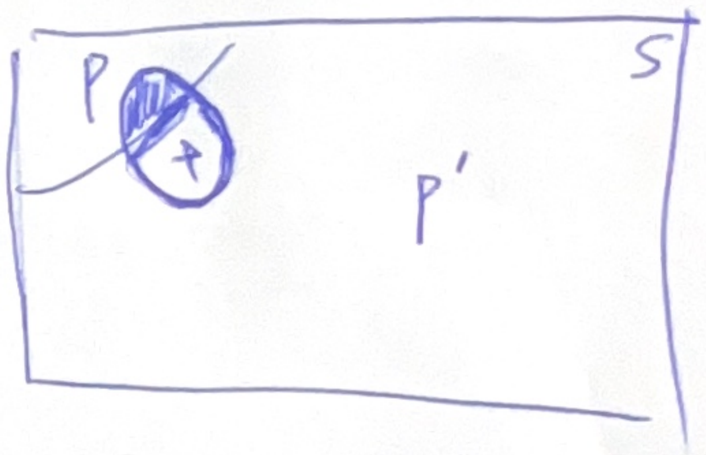
<u>Example</u>: A new method to detect pollutions in water.

P = { actually pollutant water }

P' = { not pollutant water }

$+$ = { test positive }

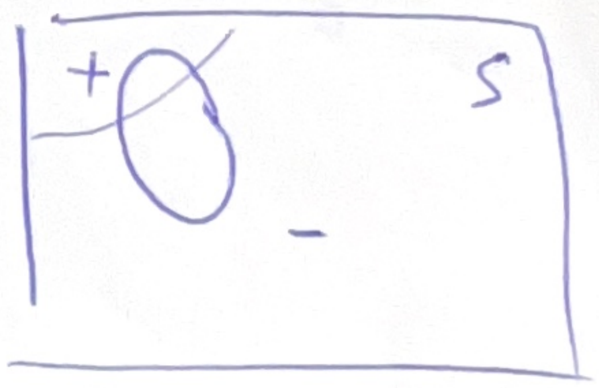
$-$ = { test negative }



$P(P) = 0.13$ in the study area.

$P(+ \mid P) = 0.91$ <u>sensitivity</u>

$P(- \mid P') = 0.94$ <u>specificity</u>



a>. Find probability of a false "+" test.

$P(+ \mid P') =$